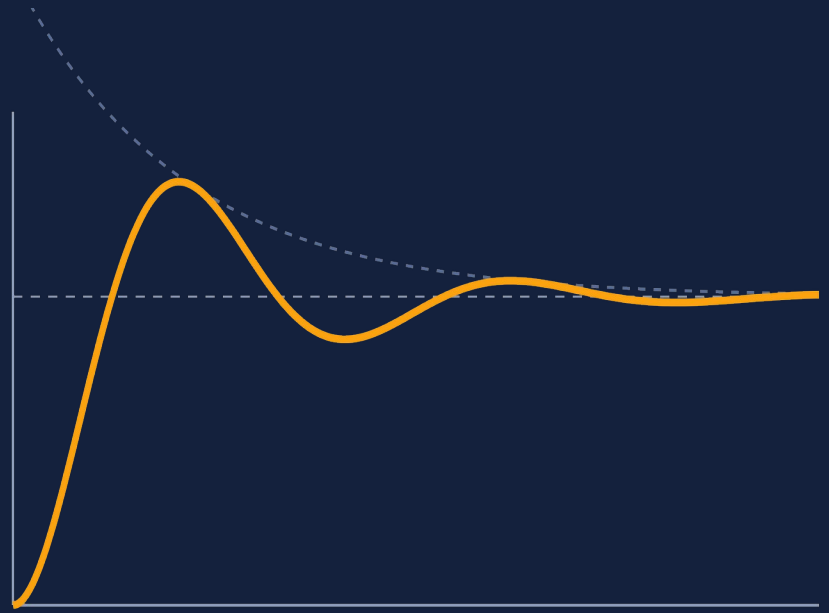
Time Response of Second-Order Systems



Control Systems (22EEG53)

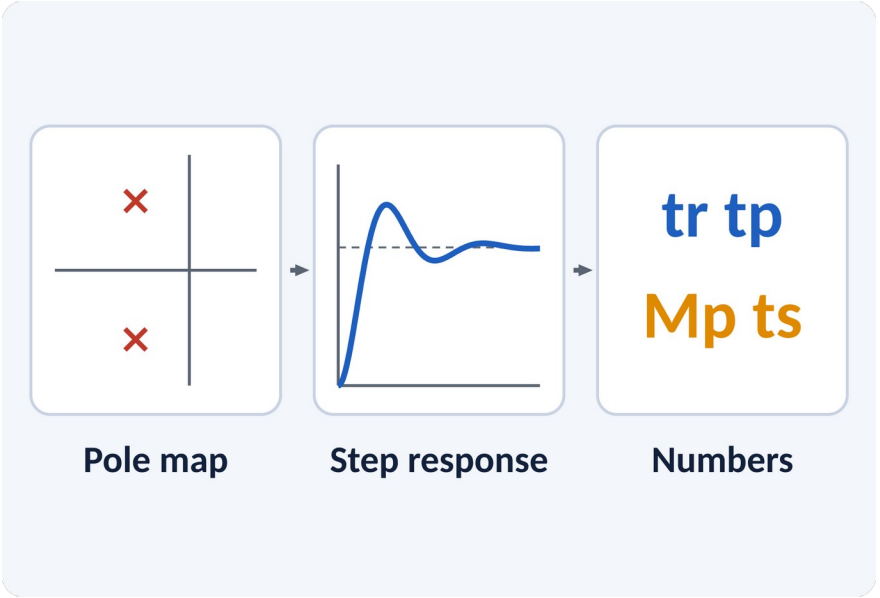
Unit 2: Time Domain Analysis of Control Systems

Topic: Second-order response and time-domain specifications

Prof. Mitesh Astik

Learning Outcomes

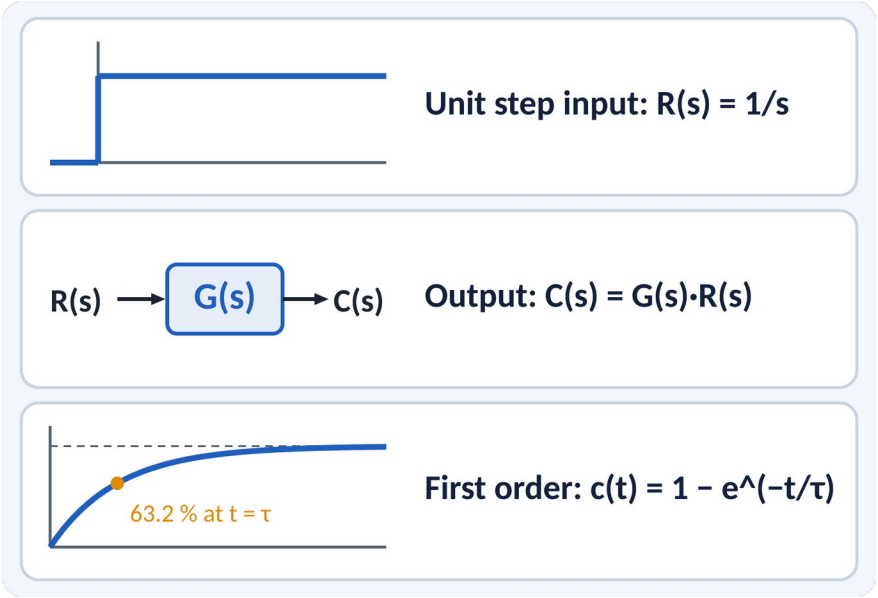
- Write the standard second-order transfer function; locate its poles.
- Classify damping and sketch the unit-step response.
- Calculate tr, tp, Mp, ts from ζ and ωn.
- Find ζ and ωn from given specifications.



Linked to CO2
Analyse first- and second-order systems' response to standard test signals.

Recap: What You Already Know

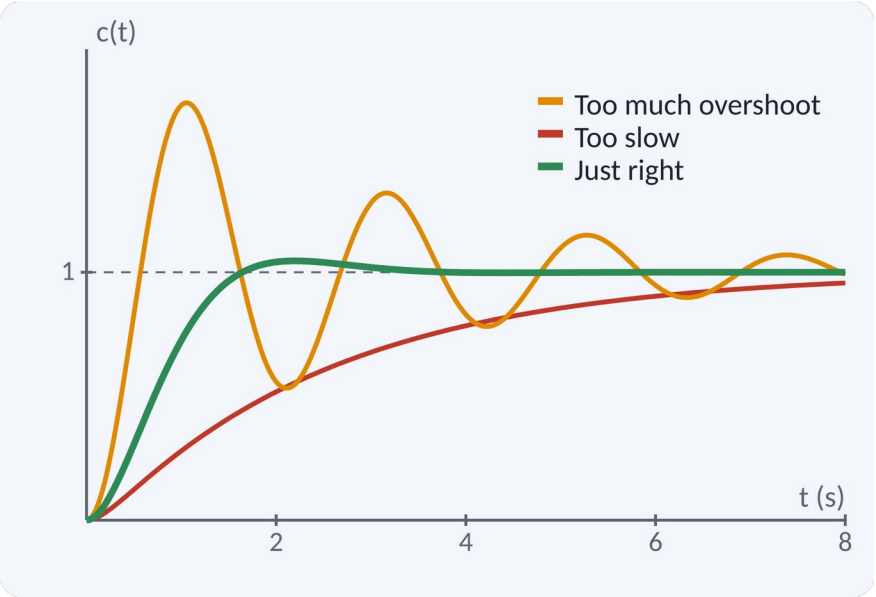
- Laplace transform of unit step is 1/s.
- Poles are roots of the denominator.
- Partial fractions give the time response.
- First-order step: c(t) = 1 − e^(−t/τ).



C(s) = G(s)·R(s)
C(s) output, G(s) transfer function, R(s) input

Motivation: Why Overshoot Matters

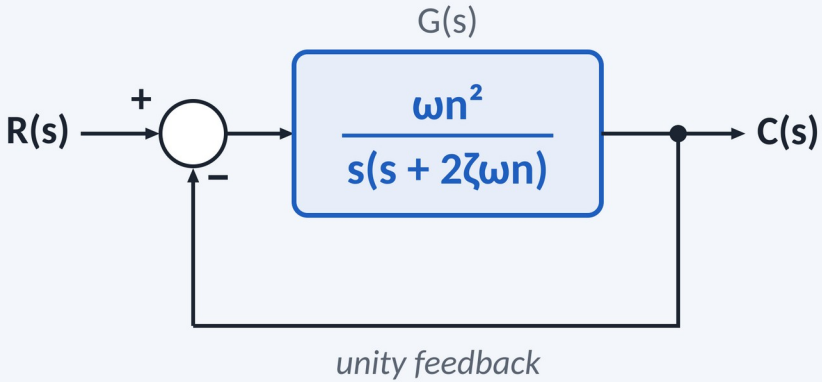
- A lift must stop at the floor without bouncing.
- A disk-drive head must settle in milliseconds.
- Antenna and robot arm: fast but not shaky.
- Lab link: DC position control (Experiment 8).
- Specs: how fast, how much overshoot, how long to settle.



Design goal: fast rise, small overshoot, short settling time

Standard Second-Order Model

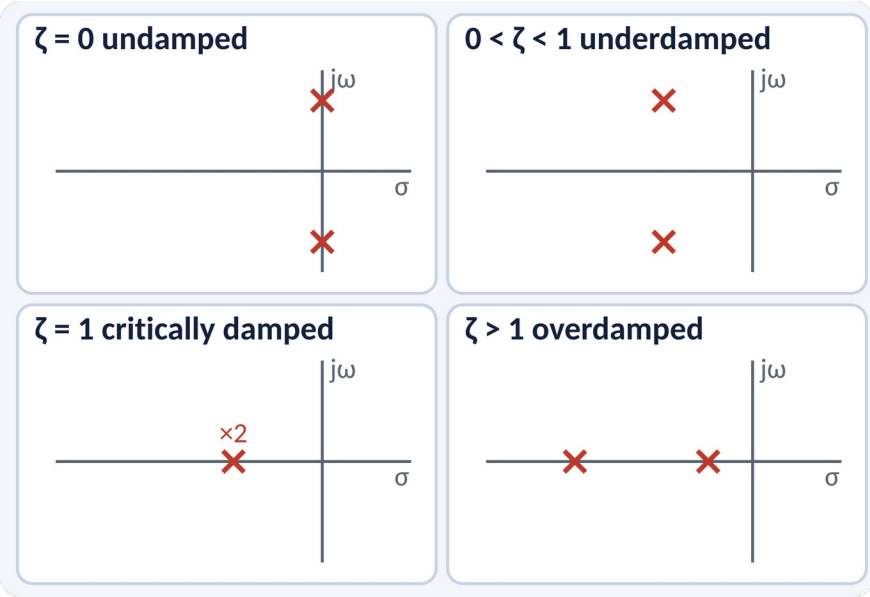
- A servo with unity feedback gives this form.
- Two parameters decide the whole response.
- ωn: how fast the system oscillates.
- ζ: how quickly oscillations die out.



$C(s)/R(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n \cdot s + \omega_n^2)$ $G(s) = \omega_n^2 / [s(s + 2\zeta\omega_n)]$
ωn natural frequency (rad/s), ζ damping ratio (no unit)

Poles and Damping Cases

- ζ = 0: poles on the imaginary axis.
- 0 < ζ < 1: complex-conjugate pair.
- ζ = 1: repeated real pole.
- ζ > 1: two distinct real poles.

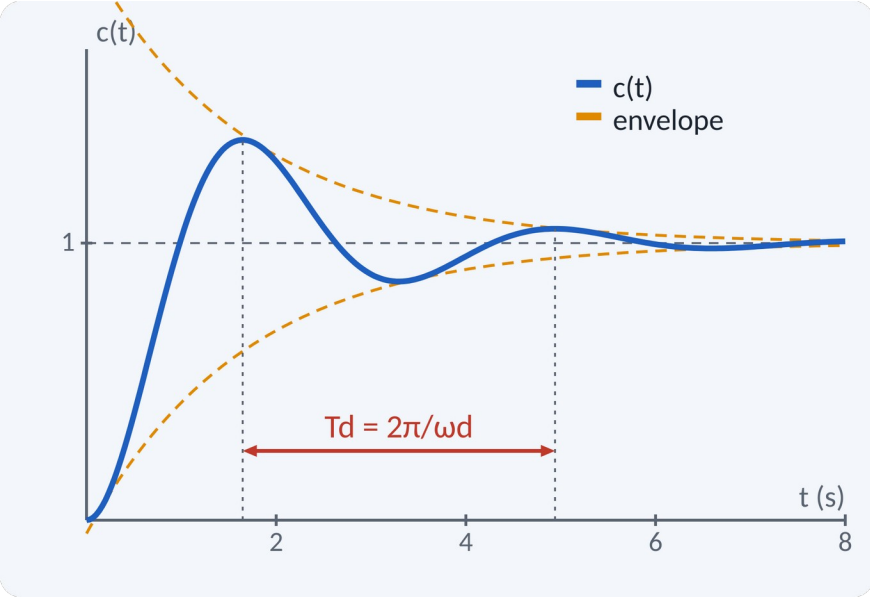


$s_{1,2} = -\zeta\omega_n \pm j\omega_d, \quad \omega_d = \omega_n\sqrt{(1-\zeta^2)}$

ω_d damped natural frequency (rad/s)

Underdamped Step Response

- Most useful case for design and exams.
- Response oscillates and settles at 1.
- Envelope decays as e^(−ζωn·t).
- Damped frequency ωd is below ωn.

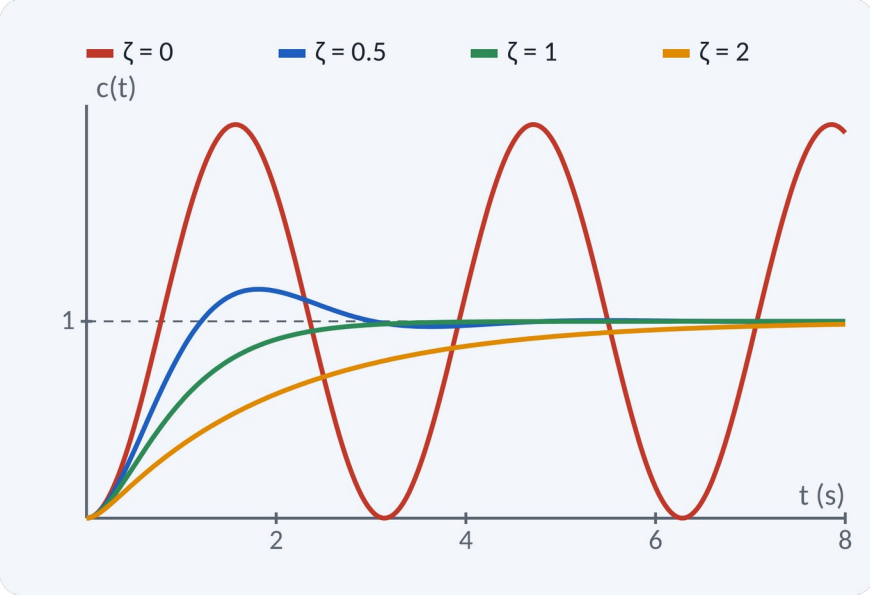


$c(t) = 1 - [e^{(-\zeta\omega_n t)}/\sqrt{(1-\zeta^2)}]\cdot\sin(\omega_d t + \varphi)$

$\varphi = \cos^{-1}\zeta$ (rad), $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$ (rad/s)

Other Damping Cases

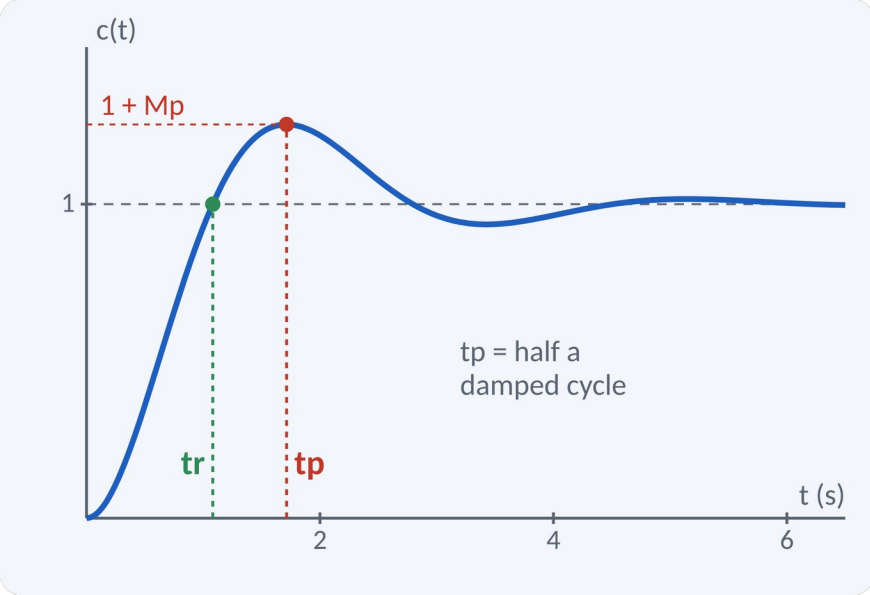
- ζ = 0: c(t) = 1 − cos ωn·t, sustained oscillation.
- ζ = 1: c(t) = 1 − (1 + ωn·t)e^(−ωn·t).
- ζ > 1: slow, no overshoot.
- Critical damping: fastest without overshoot.



Poles for ζ > 1: $s = -\zeta\omega_n \pm \omega_n\sqrt{(\zeta^2 - 1)}$

Rise Time and Peak Time

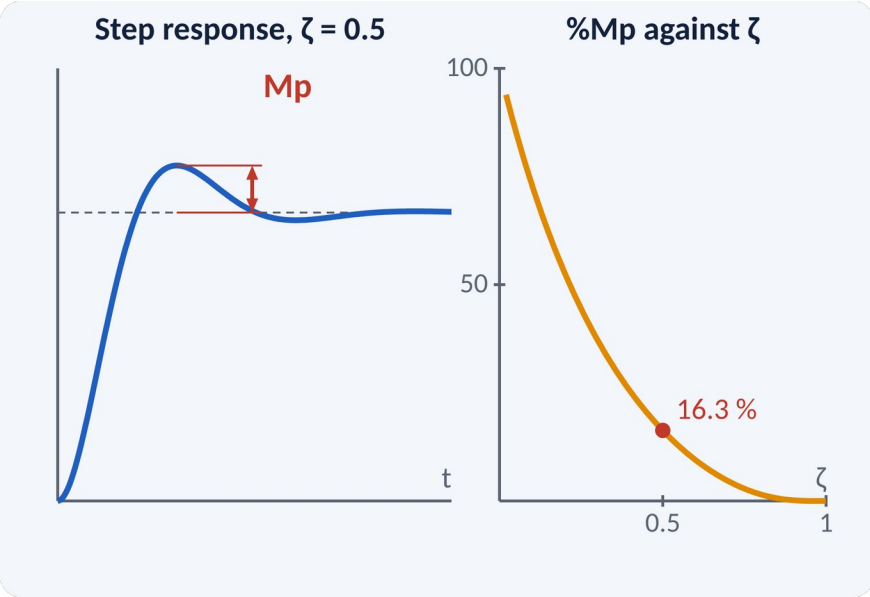
- tr: time to go from 0 to 100 % of final value.
- tp: time to reach the first peak.
- Both depend on ωd; larger ωd is faster.



$t_r = (\pi - \varphi)/\omega_d$, $\varphi = \cos^{-1}\zeta$ in rad **$t_p = \pi/\omega_d$**
t_r, t_p in s; ω_d in rad/s

Peak Overshoot

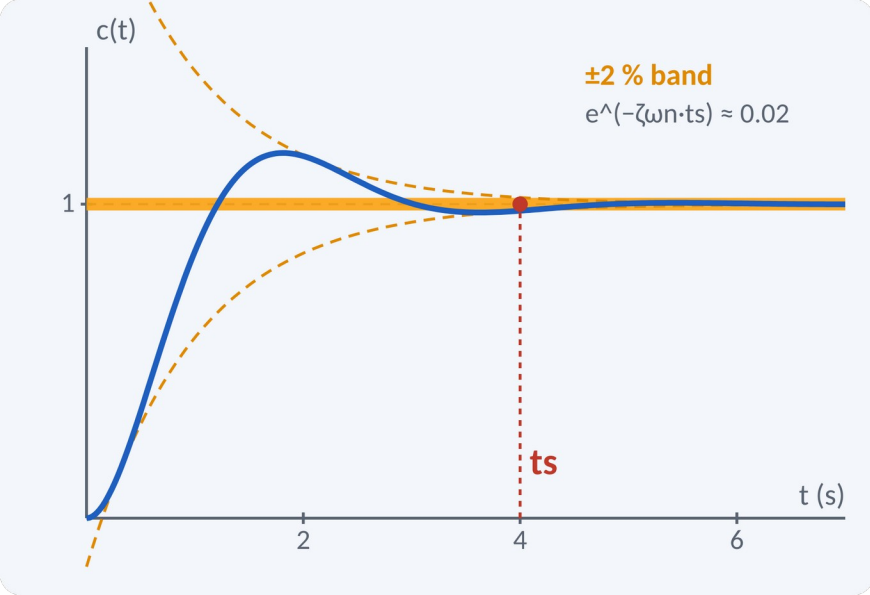
- Mp: how far the peak rises above the final value.
- Depends only on ζ, not on ωn.
- Smaller ζ gives larger overshoot.
- Typical: ζ 0.5→16.3 %, 0.6→9.5 %, 0.7→4.6 %.



Mp = e^(−πζ/√(1 − ζ²)), %Mp = 100·Mp
ζ = ln(1/Mp) / √(π² + ln²(1/Mp))

Settling Time and Delay Time

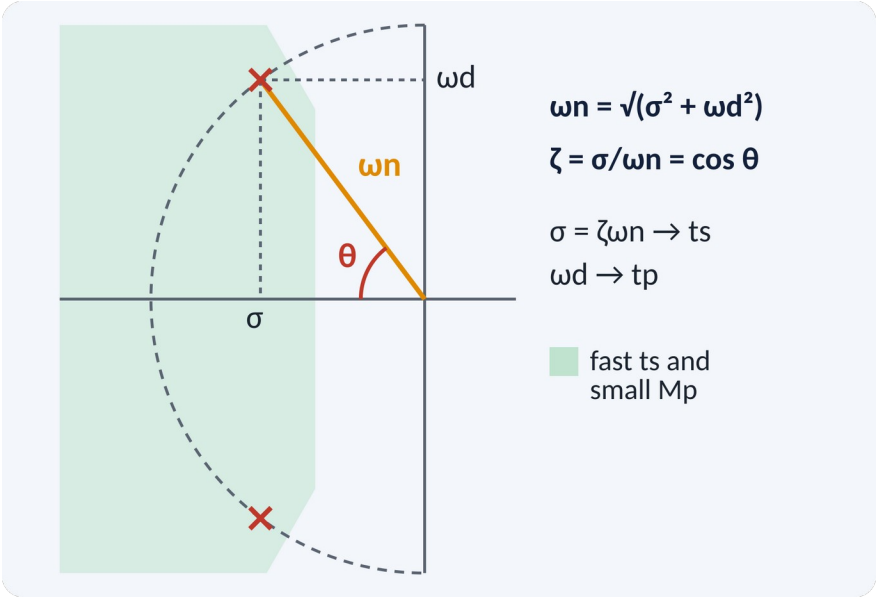
- ts: time after which response stays in a band.
- Band: ±2 % (or ±5 %) of final value.
- Set by the real part of the poles, σ = ζωn.
- td: time to reach 50 % of final value.



ts ≈ 4/(ζωn) for 2 % band; ts ≈ 3/(ζωn) for 5 % band
td ≈ (1 + 0.7ζ)/ωn

Pole Location and Specifications

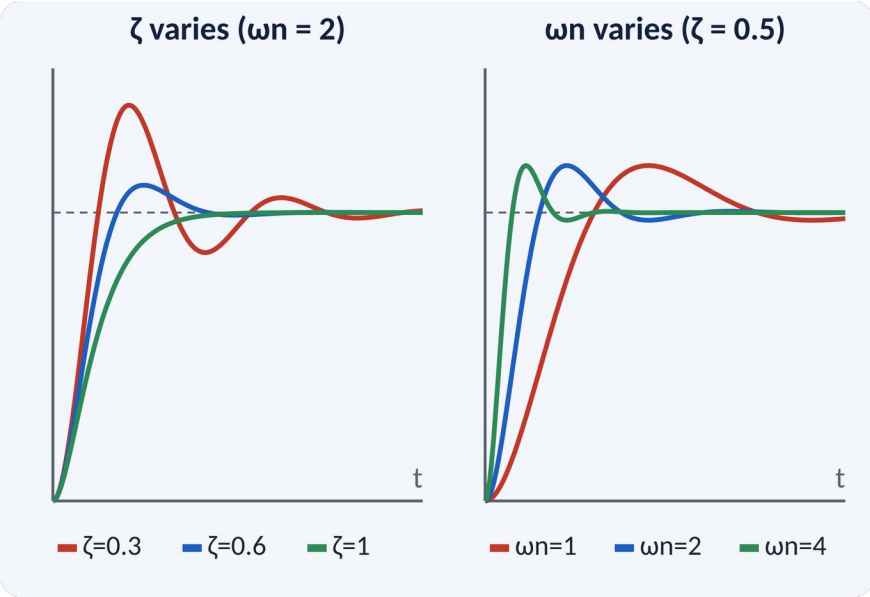
- ωn = distance of the pole from the origin.
- ζ = cos θ, θ measured from negative real axis.
- Real part σ sets settling time.
- Imaginary part ωd sets peak time.



ωn = √(σ² + ωd²); ζ = σ/ωn = cos θ

Effect of ζ and ωn

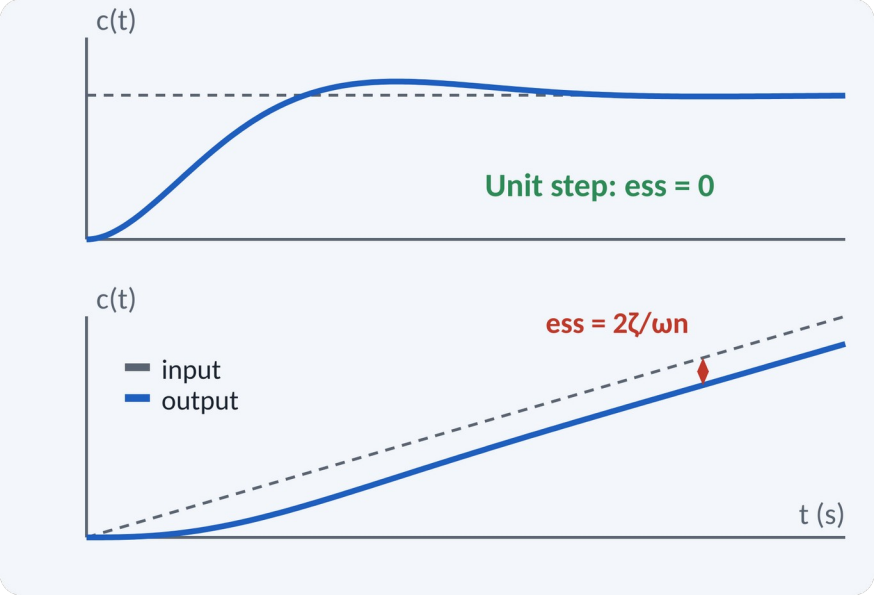
- Increase ζ: overshoot falls, peak comes later.
- Increase ωn: all times shrink; overshoot unchanged.
- Typical design choice: $0.4 \le \zeta \le 0.8$.
- Trade-off: speed against smoothness.



ζ changes the shape of the response; ωn changes its speed

Steady-State Link (Type 1)

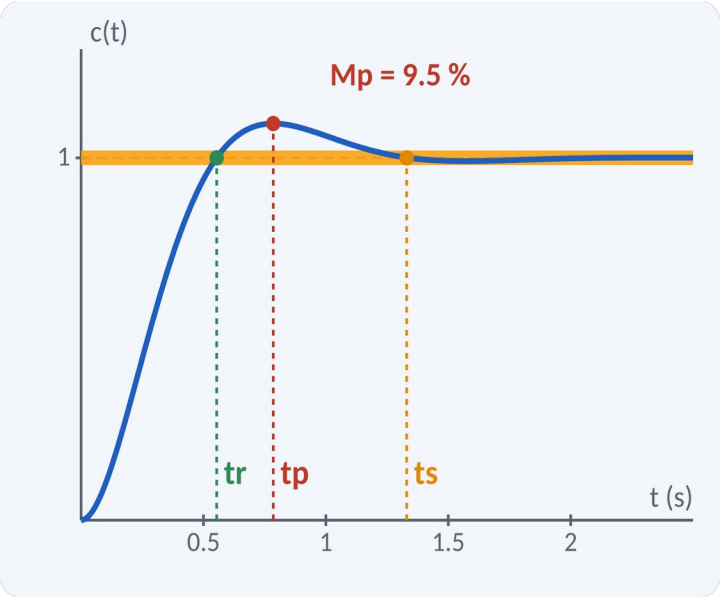
- $G(s) = \omega_n^2/[s(s + 2\zeta\omega_n)]$ is Type 1.
- Unit step input: ess = 0.
- Unit ramp input: finite error ess = 1/Kv.
- Raising ζ increases the ramp error.



$K_v = \lim_{s\to 0} s\cdot G(s) = \omega_n/(2\zeta)$; $e_{ss(ramp)} = 2\zeta/\omega_n$
K_v velocity error constant (1/s), ess steady-state error

Solved Example 1: Find the Specs

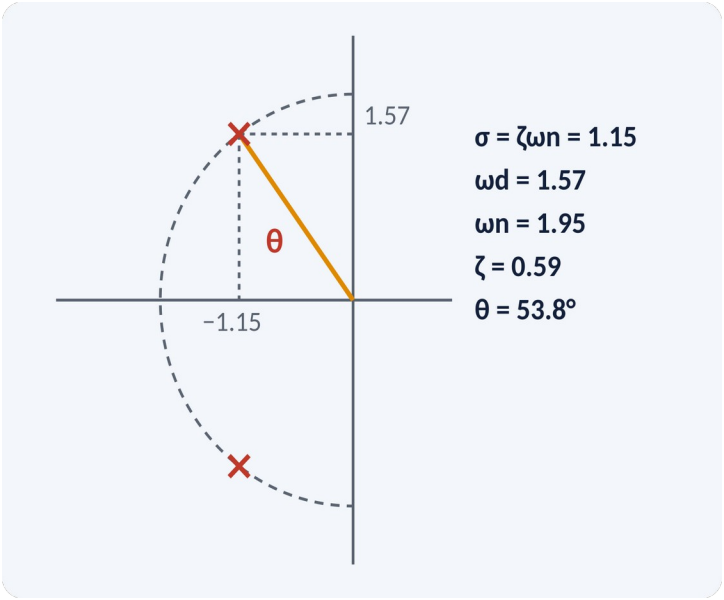
- Given G(s) = 25/[s(s + 6)], unity feedback, unit step.
- 1. T(s) = 25/(s² + 6s + 25), so ωn = 5 rad/s.
- 2. 2ζωn = 6, so ζ = 0.6; ωd = 5√0.64 = 4 rad/s.
- 3. φ = cos⁻¹0.6 = 0.927 rad; tr = (π − 0.927)/4 = 0.554 s.
- 4. tp = π/4 = 0.785 s.
- 5. Mp = e^(−π·0.6/0.8) = 0.0948, i.e. 9.5 %.
- 6. ts = 4/3 = 1.33 s (2 %); 3/3 = 1.0 s (5 %).
- 7. Kv = 25/6; ess(ramp) = 2ζ/ωn = 0.24.



Answer: ωn = 5, ζ = 0.6, tr = 0.554 s, tp = 0.785 s, Mp = 9.5 %, ts = 1.33 s

Solved Example 2: Design from Specs

- Given G(s) = K/[s(s + a)], unity feedback. Need Mp = 10 %, tp = 2 s. Find K, a, ts.
- 1. $\zeta = \ln 10/\sqrt{(\pi^2 + \ln^2 10)} = 2.303/3.895 = 0.591$.
- 2. $\omega_d = \pi/t_p = 1.571$ rad/s.
- 3. $\omega_n = \omega_d/\sqrt{(1 - \zeta^2)} = 1.571/0.807 = 1.947$ rad/s.
- 4. $K = \omega_n^2 = 3.79$.
- 5. $a = 2\zeta\omega_n = 2.30$.
- 6. $t_s = 4/(\zeta\omega_n) = 4/1.151 = 3.47$ s (2 %).



Answer: K = 3.79, a = 2.30, ts = 3.47 s; poles at −1.15 ± j1.57

Common Mistakes and Exam Tips

Common mistakes

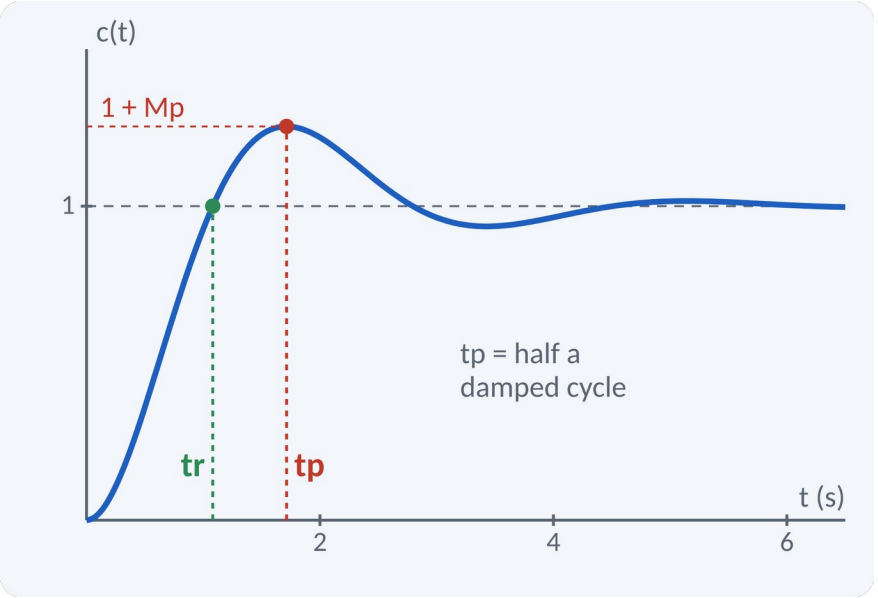
- Using ωn instead of ωd in tr and tp.
- Mixing degrees and radians for φ.
- Writing Mp as a fraction but quoting it as %.
- Quoting ts without stating the 2 % or 5 % band.
- Using the overshoot formula outside 0 < ζ < 1.

Exam tips

- Rewrite into standard form first.
- Compare coefficients to get ωn and ζ.
- Check that ζ lies between 0 and 1.
- Set the calculator to radians for φ.
- Show formula, substitution, answer with unit.

Summary

- Standard form: $\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$.
- ζ decides response type; poles $-\zeta\omega_n \pm j\omega_d$.
- $t_p = \pi/\omega_d$; $t_r = (\pi - \varphi)/\omega_d$.
- $M_p = e^{(-\pi\zeta/\sqrt{(1-\zeta^2)})}$; depends on ζ only.
- $t_s \approx 4/(\zeta\omega_n)$ for a 2 % band.



Pole position gives the specs; the specs give the pole position

Quick Quiz

Q1. A system with ζ = 0.4 is:
(a) undamped (b) underdamped (c) critically damped (d) overdamped

Ans: (b)

Q2. Peak overshoot depends only on:
(a) ωn (b) ζ (c) K only (d) ts

Ans: (b)

Q3. ζ = 0.5, ωn = 4 rad/s: ts (2 % band) =
(a) 0.5 s (b) 1 s (c) 2 s (d) 4 s

Ans: (c)

Q4. ωn = 10 rad/s, ζ = 0.6: peak time tp =
(a) 0.314 s (b) 0.393 s (c) 0.524 s (d) 0.785 s

Ans: (b)

Q5. Peak overshoot for ζ = 0.5 is about:
(a) 5 % (b) 16.3 % (c) 25 % (d) 50 %

Ans: (b)

References

1. **I. J. Nagrath and M. Gopal, Control Systems Engineering**
   5th Ed., New Age International (P) Ltd., 2008 (Unit 2 textbook)

2. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3–5 textbook)

3. **K. Ogata, Modern Control Engineering**
   5th Ed., Pearson Education Asia / PHI, 2009 (reference)

4. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

5. **Lab: MATLAB / Simulink**
   Experiments 1, 3 and 8: step response, time-domain specifications, DC position control

Next lecture: static error constants and generalized error series